

Intel® Intrinsic Guide

Updated
07/12/2024

Version
3.6.9

Instruction Set

- ☐ MMX
- ☐ SSE family
- ☐ AVX family
- ☐ AVX-512 family
- ☐ AMX family
- ☐ SVML
- ☐ Other

Categories

- ☐ Application-Targeted
- ☐ Arithmetic
- ☐ Bit Manipulation
- ☐ Cast
- ☐ Compare
- ☐ Convert
- ☐ Cryptography
- ☐ Elementary Math Functions
- ☐ General Support
- ☐ Load
- ☐ Logical
- ☐ Mask
- ☐ Miscellaneous
- ☐ Move
- ☐ OS-Targeted
- ☐ Probability/Statistics
- ☐ Random
- ☐ Set
- ☐ Shift
- ☐ Special Math Functions
- ☐ Store
- ☐ String Compare
- ☐ Swizzle
- ☐ Trigonometry
- ☐ Unknown

Release Notes

[Download: Offline Intel® Intrinsic Guide](#)

Additional resources:

- Intel® C++ Compiler Classic Developer Guide and Reference

Intel® C++ Compiler community board

All throughput and latency data is sourced from [Intel® 64 and IA-32 Architectures Software Developer Manuals](#).

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void __mm2intersect_epi32 (__m128i a, __m128i b, __mmask8* k1, __mmask8* k2)	vp2intersectd
void __mm256_2intersect_epi32 (__m256i a, __m256i b, __mmask8* k1, __mmask8* k2)	vp2intersectd
void __mm512_2intersect_epi32 (__m512i a, __m512i b, __mmask16* k1, __mmask16* k2)	vp2intersectd
void __mm2intersect_epi64 (__m128i a, __m128i b, __mmask8* k1, __mmask8* k2)	vp2intersectq
void __mm256_2intersect_epi64 (__m256i a, __m256i b, __mmask8* k1, __mmask8* k2)	vp2intersectq
void __mm512_2intersect_epi64 (__m512i a, __m512i b, __mmask8* k1, __mmask8* k2)	vp2intersectq
void __aadd_i32 (int* __A, int __B)	aadd
void __aadd_i64 (__int64* __A, __int64 __B)	aadd
void __aand_i32 (int* __A, int __B)	aand
void __aand_i64 (__int64* __A, __int64 __B)	aand
__m128i __mm_abs_epi16 (__m128i a)	pabsw
__m128i __mm_mask_abs_epi16 (__m128i src, __mmask8 k, __m128i a)	vpabsw
__m128i __mm_maskz_abs_epi16 (__mmask8 k, __m128i a)	vpabsw
__m256i __mm256_abs_epi16 (__m256i a)	vpabsw
__m256i __mm256_mask_abs_epi16 (__m256i src, __mmask16 k, __m256i a)	vpabsw
__m256i __mm256_maskz_abs_epi16 (__mmask16 k, __m256i a)	vpabsw
__m512i __mm512_abs_epi16 (__m512i a)	vpabsw
__m512i __mm512_mask_abs_epi16 (__m512i src, __mmask32 k, __m512i a)	vpabsw
__m512i __mm512_maskz_abs_epi16 (__mmask32 k, __m512i a)	vpabsw
__m128i __mm_abs_epi32 (__m128i a)	pabsd
__m128i __mm_mask_abs_epi32 (__m128i src, __mmask8 k, __m128i a)	vpabsd
__m128i __mm_maskz_abs_epi32 (__mmask8 k, __m128i a)	vpabsd
__m256i __mm256_abs_epi32 (__m256i a)	vpabsd
__m256i __mm256_mask_abs_epi32 (__m256i src, __mmask8 k, __m256i a)	vpabsd
__m256i __mm256_maskz_abs_epi32 (__mmask8 k, __m256i a)	vpabsd
__m512i __mm512_abs_epi32 (__m512i a)	vpabsd
__m512i __mm512_mask_abs_epi32 (__m512i src, __mmask16 k, __m512i a)	vpabsd
__m512i __mm512_maskz_abs_epi32 (__mmask16 k, __m512i a)	vpabsd
__m128i __mm_abs_epi64 (__m128i a)	vpabsq

__m128i __mm_mask_abs_epi64 (__m128i src,	vpabsq
__mmask8 k, __m128i a)	
__m128i __mm_maskz_abs_epi64 (__mmask8 k,	vpabsq
__m128i a)	
__m256i __mm256_abs_epi64 (__m256i a)	vpabsq
__m256i __mm256_mask_abs_epi64 (__m256i	vpabsq
src, __mmask8 k, __m256i a)	
__m256i __mm256_maskz_abs_epi64 (__mmask8	vpabsq
k, __m256i a)	
__m512i __mm512_abs_epi64 (__m512i a)	vpabsq
__m512i __mm512_mask_abs_epi64 (__m512i	vpabsq
src, __mmask8 k, __m512i a)	
__m512i __mm512_maskz_abs_epi64 (__mmask8	vpabsq
k, __m512i a)	
__m128i __mm_abs_epi8 (__m128i a)	pabsb
__m128i __mm_mask_abs_epi8 (__m128i src,	vpabsb
__mmask16 k, __m128i a)	
__m128i __mm_maskz_abs_epi8 (__mmask16 k,	vpabsb
__m128i a)	
__m256i __mm256_abs_epi8 (__m256i a)	vpabsb
__m256i __mm256_mask_abs_epi8 (__m256i src,	vpabsb
__mmask32 k, __m256i a)	
__m256i __mm256_maskz_abs_epi8 (__mmask32	vpabsb
k, __m256i a)	
__m512i __mm512_abs_epi8 (__m512i a)	vpabsb
__m512i __mm512_mask_abs_epi8 (__m512i src,	vpabsb
__mmask64 k, __m512i a)	
__m512i __mm512_maskz_abs_epi8 (__mmask64	vpabsb
k, __m512i a)	
__m512d __mm512_abs_pd (__m512d v2)	vpandq
__m512d __mm512_mask_abs_pd (__m512d src,	vpandq
__mmask8 k, __m512d v2)	
__m128h __mm_abs_ph (__m128h v2)	...
__m256h __mm256_abs_ph (__m256h v2)	...
__m512h __mm512_abs_ph (__m512h v2)	...
__m64 __mm_abs_pi16 (__m64 a)	pabsw
__m64 __mm_abs_pi32 (__m64 a)	pabsd
__m64 __mm_abs_pi8 (__m64 a)	pabsb
__m512 __mm512_abs_ps (__m512 v2)	vpandd
__m512 __mm512_mask_abs_ps (__m512 src,	vpandd
__mmask16 k, __m512 v2)	
__m128d __mm_acos_pd (__m128d a)	...
__m256d __mm256_acos_pd (__m256d a)	...
__m512d __mm512_acos_pd (__m512d a)	...
__m512d __mm512_mask_acos_pd (__m512d src,	...
__mmask8 k, __m512d a)	
__m128h __mm_acos_ph (__m128h a)	...
__m256h __mm256_acos_ph (__m256h a)	...
__m512h __mm512_acos_ph (__m512h a)	...
__m512h __mm512_mask_acos_ph (__m512h src,	...
__mmask32 k, __m512h a)	
__m128 __mm_acos_ps (__m128 a)	...
__m256 __mm256_acos_ps (__m256 a)	...
__m512 __mm512_acos_ps (__m512 a)	...
__m512 __mm512_mask_acos_ps (__m512 src,	...
__mmask16 k, __m512 a)	
__m128d __mm_acosh_pd (__m128d a)	...
__m256d __mm256_acosh_pd (__m256d a)	...
__m512d __mm512_acosh_pd (__m512d a)	...
__m512d __mm512_mask_acosh_pd (__m512d src,	...
__mmask8 k, __m512d a)	
__m128h __mm_acosh_ph (__m128h a)	...
__m256h __mm256_acosh_ph (__m256h a)	...
__m512h __mm512_acosh_ph (__m512h a)	...

__m512h	_mm512_mask_acosh_ph	(__m512h src,	...
__mmask32	k,	__m512h a)	
__m128	_mm_acosh_ps	(__m128 a)	...
__m256	_mm256_acosh_ps	(__m256 a)	...
__m512	_mm512_acosh_ps	(__m512 a)	...
__m512	_mm512_mask_acosh_ps	(__m512 src,	...
__mmask16	k,	__m512 a)	
__m128i	_mm_add_epi16	(__m128i a, __m128i	paddw
	b)		
__m128i	_mm_mask_add_epi16	(__m128i src,	vpaddw
__mmask8	k,	__m128i a, __m128i b)	
__m128i	_mm_maskz_add_epi16	(__mmask8 k,	vpaddw
__m128i	a, __m128i b)		
__m256i	_mm256_add_epi16	(__m256i a,	vpaddw
__m256i	b)		
__m256i	_mm256_mask_add_epi16	(__m256i	vpaddw
src,	__mmask16 k,	__m256i a, __m256i b)	
__m256i	_mm256_maskz_add_epi16	(__mmask16	vpaddw
k,	__m256i a, __m256i b)		
__m512i	_mm512_add_epi16	(__m512i a,	vpaddw
__m512i	b)		

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